Midterm 2 W26

2026-03-03

Instructions

This is an in-person exam. Your exam must be complete and uploaded to Gradescope by 2:00p.

Answer the following questions and complete the exercises in RMarkdown. Please embed all of your code and push your final work to your repository. Your code must be organized, clean, and run free from errors. Remember, you must remove the `#` for any included code chunks to run. Be sure to add your name to the author header above.

Your code must knit in order to be considered. If you are stuck and cannot answer a question, then comment out your code and knit the document. You may use your notes, labs, and homework to help you complete this exam. Do not use any other resources- including AI assistance or other students' work.

Don't forget to answer any questions that are asked in the prompt! Each question must be coded; it cannot be answered by a sort in a spreadsheet or a written response only.

For all plots you create, a title and clearly labeled axes must be provided.

Be sure to push your completed midterm to your repository and upload the document to Gradescope. *Incorrectly uploading the exam to Gradescope will result in a 5 point deduction.* This exam is worth 50 points.

Load the libraries

```
library(tidyverse)
library(janitor)
library(naniar)
library(ggmap)
library(lubridate)
library(shiny)
```

API Key

Since you will build a map as part of the exam, take a moment to register your API key. You may need to look up your stadia maps (https://client.stadiamaps.com/dashboard/overview) API key online (please do that now).

```
register_stadiamaps("d3a65b7a-5eac-457a-a807-159dd3e22426", write = FALSE)
```

About the data

This study is focused on the fasting behavior of polar bears in the Southern Beaufort (SB) and Chukchi seas (CS). The data were collected from 1983-2017 and include measurements of blood urea nitrogen (BUN) and creatinine, which are used to calculate a ratio that indicates whether a bear is fasting or not. The original paper is included in the data folder. *It may help to quickly read the abstract before starting the exam.*

The data are from: Rode, K.D., Wilson, R.R., Douglas, D.C., Muhlenbruch, V., Atwood, T.C., Regehr, E.V., Richardson, E.S., Pilfold, N.W., Derocher, A.E., Durner, G.M., Stirling, I., Amstrup, S.C., St. Martin, M., Pagano, A.M., and others, 2017, Spring fasting behavior in a marine apex predator provides an index of ecosystem productivity: Global Change Biology, v. 24, no. 1, p. 410–423, https://doi.org/10.1111/gcb.13933 (https://doi.org/10.1111/gcb.13933).

Data resources

- https://data.usgs.gov/datacatalog/data/USGS:ASC141 (https://data.usgs.gov/datacatalog/data/USGS:ASC141)
- https://www.sciencebase.gov/catalog/item/61a1a9e7d34eb622f69743d2 (https://www.sciencebase.gov/catalog/item/61a1a9e7d34eb622f69743d2)

The variables include:

- `bear_id` : A unique number given to each bear.
- `capture_date` : The data when the bear was tranquilized and assessed.
- `popn` : The population to which the bear belongs.
- `sex` : Sex of each bear.
- `age_class` : The age class of each bear (A=adult, S=subadult).
- `mating` : Whether or not the bear is pursuing a mate.
- `denning` : Whether or not the bear denned last year.
- `lat` : Latitude where the bear was located.
- `long` : Longitude where the bear was located.
- `urea_nitrogen` : Concentration of urea nitrogen.
- `creatinine` : Concentration of creatinine.

Load the data

Run the code chunk below to load the data, clean the names, and do some tidying.

```
polar_bears <-
  read_csv("data/polarBear_serumUreaCreatinine_beaufortChukchi_rode.csv") %>%
  clean_names() %>%
  select(-den_method, -no_cubs, -cub_age) %>%
  filter(age_class==c("A", "S")) %>%
  mutate(sex=toupper(sex),
         age_class=toupper(age_class))
```

1. (2 points) List the variable names and use a summary function to get an idea of the data structure.

```
names(polar_bears)
```

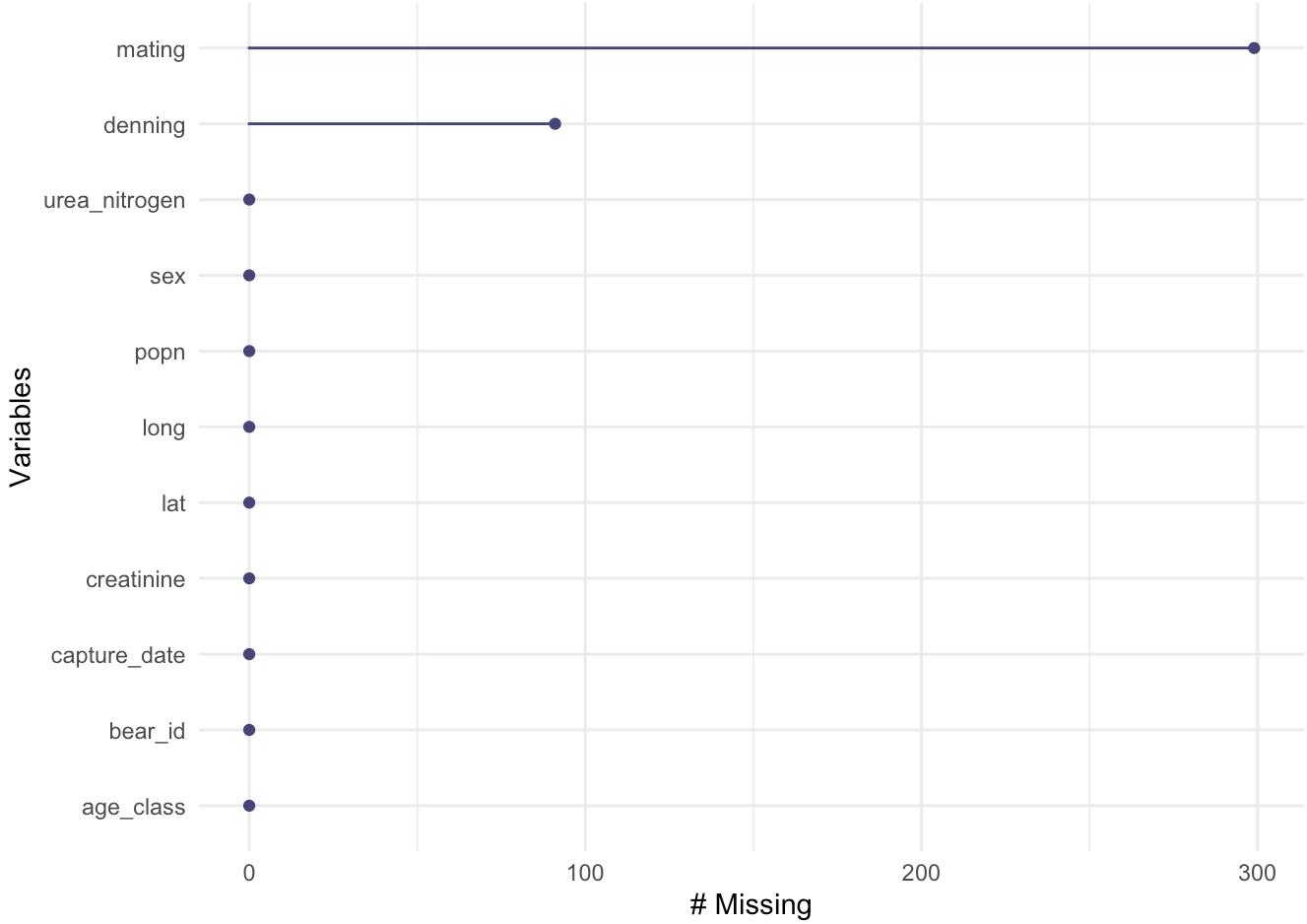
```
##  [1] "bear_id"       "capture_date"  "popn"          "sex"
##  [5] "age_class"     "mating"        "denning"       "lat"
##  [9] "long"          "urea_nitrogen" "creatinine"
```

```
summary(polar_bears)
```

```
##     bear_id       capture_date            popn               sex
##  Min.   :  697   Min.   :1983-04-23   Length:519         Length:519
##  1st Qu.:20166   1st Qu.:2000-04-10   Class :character   Class :character
##  Median :20705   Median :2008-04-12   Mode  :character   Mode  :character
##  Mean   :18040   Mean   :2004-07-02
##  3rd Qu.:21220   3rd Qu.:2011-03-31
##  Max.   :99990   Max.   :2016-04-22
##   age_class            mating            denning               lat
##  Length:519         Length:519         Length:519         Min.   :62.46
##  Class :character   Class :character   Class :character   1st Qu.:68.11
##  Mode  :character   Mode  :character   Mode  :character   Median :70.53
##                                                           Mean   :69.86
##                                                           3rd Qu.:71.00
##                                                           Max.   :72.70
##       long        urea_nitrogen     creatinine
##  Min.   :-172.2   Min.   : 1.00   Min.   :0.350
##  1st Qu.:-165.2   1st Qu.: 6.00   1st Qu.:0.850
##  Median :-150.7   Median :11.00   Median :1.050
##  Mean   :-153.7   Mean   :15.13   Mean   :1.079
##  3rd Qu.:-146.5   3rd Qu.:21.25   3rd Qu.:1.285
##  Max.   :-134.8   Max.   :69.00   Max.   :2.350
```

2. (3 points) How are NA's represented and where are they distributed in the variables? Use the `naniar` package to visualize the distribution of NA's in the data and replace any alternative representations of NA with actual NA values.

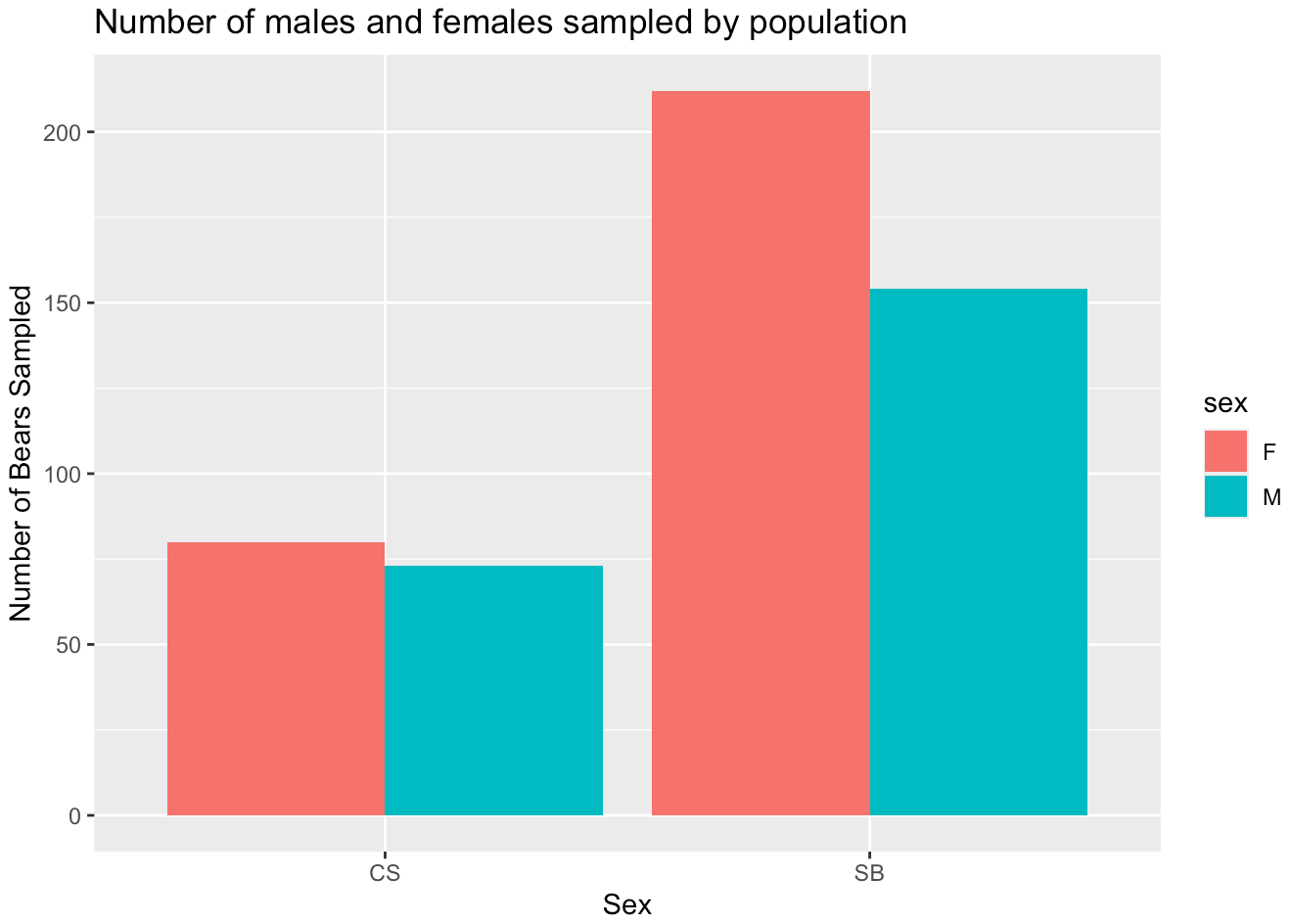
```
polar_bears <- polar_bears %>%
  replace_with_na(replace = list(mating="unknown",
                                 denning="unknown"))

polar_bears %>%
  gg_miss_var()
```



There are some “unknown” in column “mating” and “denning”, and there is no other NA’s in other variables

3. (6 points) Is sampling even for both sexes? Make a count the shows the number of males and females sampled by population and make a plot to visualize the counts.

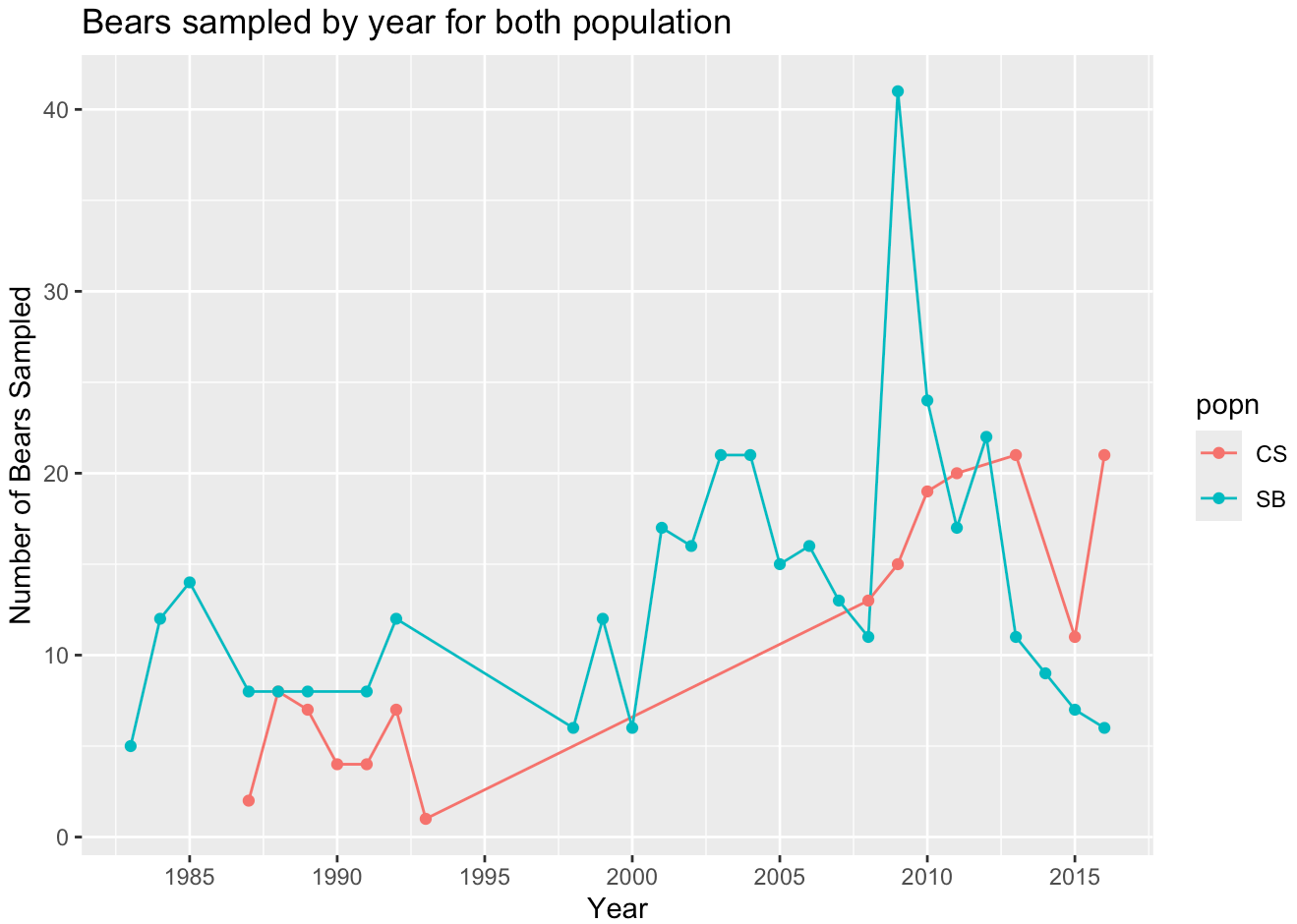
```
polar_bears %>%
  ggplot(aes(x=popn,fill = sex))+
  geom_bar(position="dodge")+
  labs(x="Sex",y="Number of Bears Sampled",title="Number of males and females sampled by population")
```



For both population, there are more female sampled. It is almost even for CS population, but not even sampling for SB population.

4. (6 points) How many bears were sampled in each year for both populations? Make a line plot(s) to visualize sampling by year for both populations. (hint: you will need to pull apart the date to get the year in its own column)

```
polar_bears <- polar_bears %>%
  mutate(capture_year=year(ymd(capture_date))) %>%
  mutate(capture_year=as.numeric(capture_year))

polar_bears %>%
  group_by(capture_year,popn) %>%
  count() %>%
  ggplot(aes(x=capture_year,y=n,color=popn))+
  geom_line()+
  geom_point()+
  scale_x_continuous(breaks=seq(1980,2020,by=5))+
  labs(x="Year",y="Number of Bears Sampled",title="Bears sampled by year for both popula
tion")
```



5. (2 points) What are the min, median, mean, and max values for `urea_nitrogen` and `creatinine`?

```
polar_bears %>%
  summarise(min_urea=min(urea_nitrogen),median_urea=median(urea_nitrogen),mean_urea=mean
(urea_nitrogen),max_urea=max(urea_nitrogen))
```

```
## # A tibble: 1 × 4
##   min_urea median_urea mean_urea max_urea
##      <dbl>       <dbl>     <dbl>    <dbl>
## 1        1          11      15.1       69
```

```
polar_bears %>%
  summarise(min_creatinine=min(creatinine),median_creatinine=median(creatinine),mean_cre
atinine=mean(creatinine),max_creatinine=max(creatinine))
```

```
## # A tibble: 1 × 4
##   min_creatinine median_creatinine mean_creatinine max_creatinine
##            <dbl>             <dbl>           <dbl>          <dbl>
## 1           0.35              1.05            1.08           2.35
```

6. (3 points) As explained above, the ratio of `urea_nitrogen` to `creatinine` is used as an index of fasting. To calculate the ratio, we need to multiply `urea_nitrogen` by 0.466, then divide by `creatinine`. Create a new variable called `un_creat_ratio` that contains this calculation.

```
polar_bears <- polar_bears %>%
  mutate(un_creat_ratio=urea_nitrogen*0.466/creatinine)
polar_bears %>%
  select(bear_id,urea_nitrogen,creatinine,un_creat_ratio)
```

```
## # A tibble: 519 × 4
##    bear_id urea_nitrogen creatinine un_creat_ratio
##      <dbl>         <dbl>      <dbl>          <dbl>
##  1    6398          12.3       0.57         10.1
##  2    6384           3         1.45          0.964
##  3    6520          30         0.85         16.4
##  4    6559          18         1.1           7.63
##  5   20629          10         1             4.66
##  6   20876           5         1.3           1.79
##  7   20354          29.5       0.65         21.1
##  8   21024          29         0.7          19.3
##  9   20764          19.5       1.2           7.57
## 10   21045          29         0.5          27.0
## # ℹ 509 more rows
```

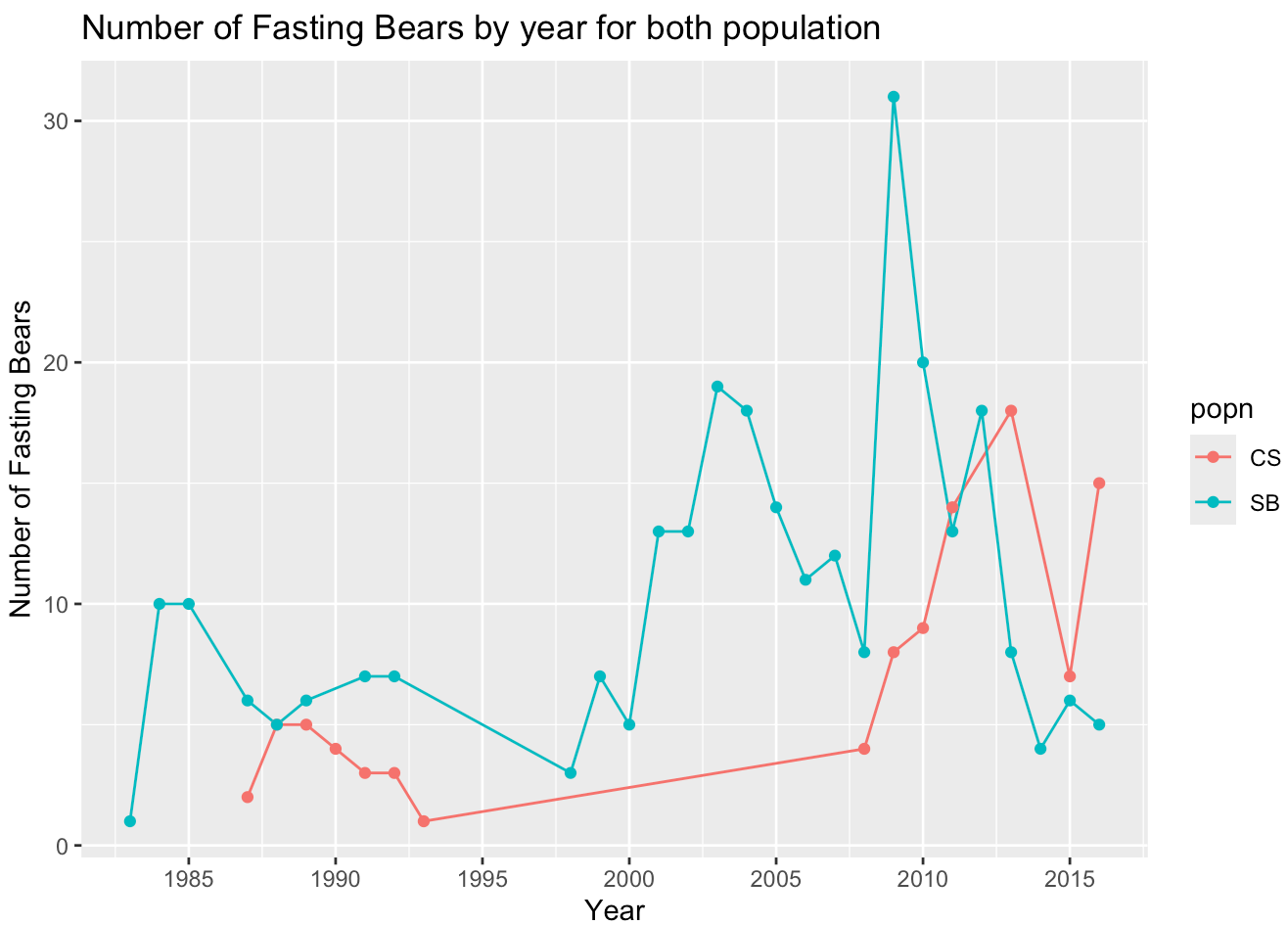
7. (4 points) The authors of the study categorized bears as fasting if their `un_creat_ratio` was less than or equal to 10. Use `case_when()` to create a new variable called `fasting` that categorizes bears as fasting or not.

```
polar_bears <- polar_bears %>%
  mutate(fasting=case_when(
    un_creat_ratio <= 10 ~ "fasting",
    un_creat_ratio > 10 ~ "not fasting"
  ))
polar_bears %>%
  select(bear_id,un_creat_ratio,fasting)
```

```
## # A tibble: 519 × 3
##    bear_id un_creat_ratio fasting
##      <dbl>          <dbl> <chr>
##  1    6398         10.1   not fasting
##  2    6384          0.964 fasting
##  3    6520         16.4   not fasting
##  4    6559          7.63  fasting
##  5   20629          4.66  fasting
##  6   20876          1.79  fasting
##  7   20354         21.1   not fasting
##  8   21024         19.3   not fasting
##  9   20764          7.57  fasting
## 10   21045         27.0   not fasting
## # ℹ 509 more rows
```

8. (6 points) The authors found that the number of fasting bears increased in the SB population but decreased in the CS population. Make a plot(s) that shows the number of fasting bears by year for both populations to visualize this trend.

```
polar_bears %>%
  filter(fasting=="fasting") %>%
  group_by(capture_year,popn) %>%
  count() %>%
  ggplot(aes(x=capture_year,y=n,color=popn))+
  geom_line()+
  geom_point()+
  scale_x_continuous(breaks=seq(1980,2020,by=5))+
  labs(x="Year",y="Number of Fasting Bears",title="Number of Fasting Bears by year for b
oth population")
```



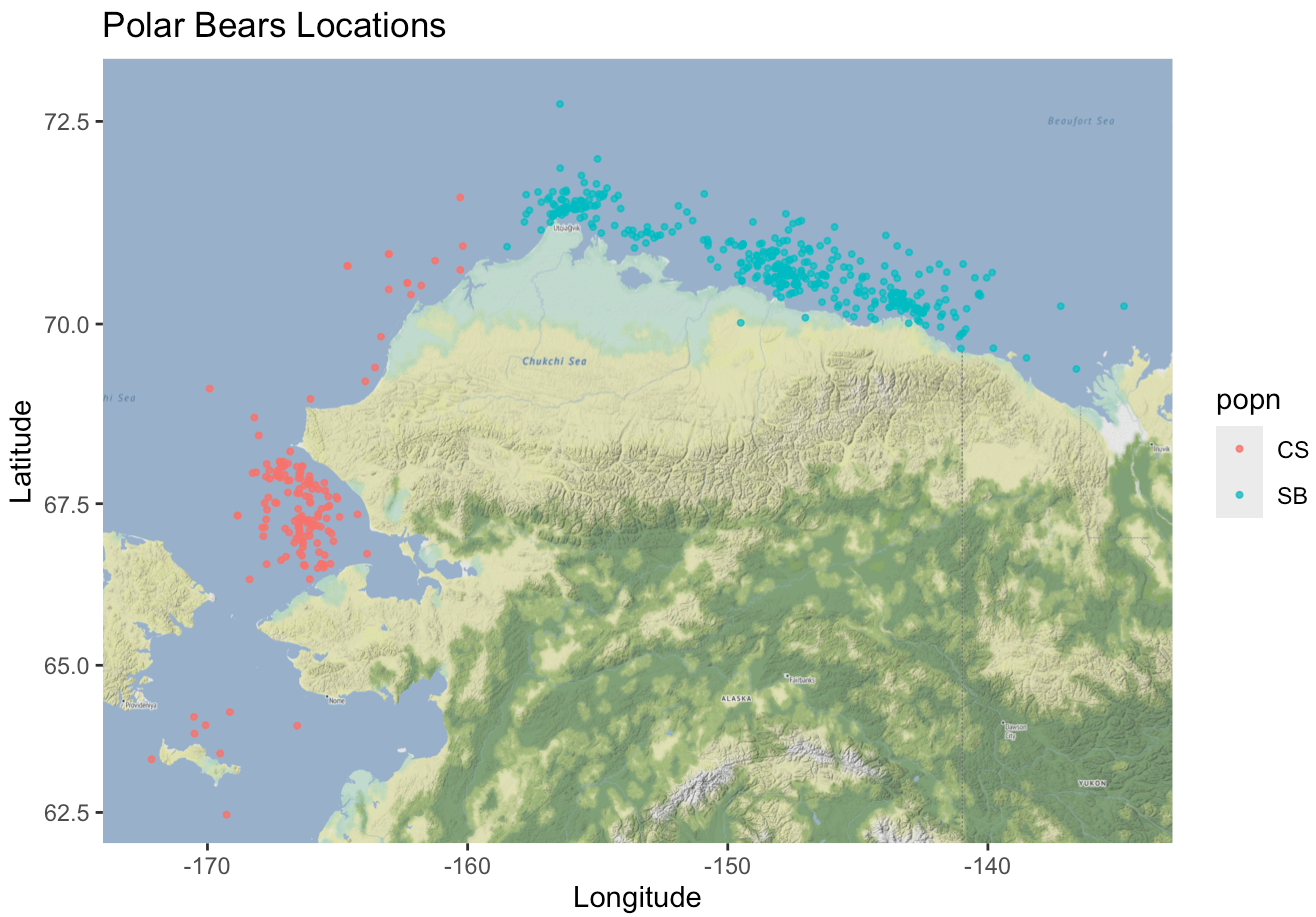
9. (8 points) Where are the polar bears used in this study located? Make a static (not leaflet) distribution map that shows the locations of bears by population. (hint: use `color=popn` in the `aes()` of `geom_point()`)

```
lon_min=min(polar_bears$long,na.rm=T)
lon_max=max(polar_bears$long,na.rm=T)
lat_min=min(polar_bears$lat,na.rm=T)
lat_max=max(polar_bears$lat,na.rm=T)

bbox = make_bbox(c(lon_min,lon_max),c(lat_min,lat_max))

map1<- get_stadiamap(bbox,maptype = "stamen_terrain",zoom=6)
```

```
## ℹ © Stadia Maps © Stamen Design © OpenMapTiles © OpenStreetMap contributors.
```

```
## ℹ 48 tiles needed, this may take a while (try a smaller zoom?)
```

```
ggmap(map1)+
  geom_point(data=polar_bears,mapping=aes(x=long,y=lat,color=popn),size=0.75,alpha=0.8)+
  labs(x="Longitude",y="Latitude",title="Polar Bears Locations")
```



10. (10 points) Build a simple shiny app that shows the range of `un_creat_ratio` values for both populations. The user should be able to select the fill variable and choose between sex, mating, age_class, denning, and fasting. Please use the code chunk below so that your exam knits.

```
ui <- fluidPage(
  selectInput("fill",
              "Select fill variable",
              choices = c("sex", "mating", "age_class", "denning", "fasting"),
              selected = "sex"),
  plotOutput("plot")
)

server <- function(input, output, session) {
  output$plot = renderPlot({
    polar_bears %>%
      na.omit() %>%
      ggplot(aes(x=popn,y=un_creat_ratio,fill=.data[[input$fill]]))+
      geom_boxplot(na.rm=T)+
      labs(x="Population",y="urea nitrogen to creatinine ratio",title = "Range of urea n
itrogen to creatinine ratio for both population")
  })
}

shinyApp(ui, server)
```

Submit the Midterm

1. Save your work and knit the .rmd file.
2. Open the .html file and “print” it to a .pdf file in Google Chrome (not Safari).
3. Go to the class Canvas page and open Gradescope.
4. Submit your .pdf file to the midterm assignment- be sure to assign the pages to the correct questions.
5. Commit and push your work to your repository.